CO I A - 1

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1)

A hacker is a person who uses technical knowledge and computer skills to identify weaknesses, access computer systems, or modify information

Types of Hacker

1. White Hat: works legally with permission to find security weakness.

2. Black Hat: Accesses systems illegally to steal data or cause damage

3. Grey Hat: Accesses system without permission but may not have harmful intentions.

4) Script kiddie: uses ready-made hacking tools without technical knowledge

5) Hacktivist: Hacks system for political or social causes

6) Insider: misuses access given by an organization.

Example:

=) Hired testers - White Hat

=) Former employees - Insider

=) Student using downloaded tools - Script kiddie.

=) Activist group - Hacktivist

Ethical Principle

1) Obtain written Permission
2) Maintain confidentially
3) Avoid damaging systems.

2) Ans

The information Technology Act, 2000 Protects computer systems and data from unauthorized access.

Important sections:

1. section 43: Deals with unauthorized access, downloading and copying data.
2. section 66: covers dishonest or fraudulent computer-related acts.
3. section 72: Deals with breach of confidentially and Privacy in specified circumstances.

Types of Disclosure

1. Full Disclosure: Publicly reveals the vulnerability details.
2. Responsible Disclosure: Privately informs the organization and allows time to fix problem.

Bug Bounty: Researchers report vulnerabilities through an official programme and may receive rewards.

Recommended method:

=) stop further testing
=) Do not Download or share student data.
=) Report the vulnerability Privately.
=) Give the university time to fix it.

3) Aus

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Ethical hacking is a systematic Process of identifying security weaknesses

Phases of Ethical Hacking

1) Reconnaissance: collect information about the target
2) scanning: identify open Ports, services and vulnerabilities.
3) Gaining Access: Test vulnerabilities to check system security.
4) maintaining Access: check wheather access can be maintained.
5) Reporting: Prepare a report with vulnerabilities and solutions.

Rules of Engagement:

1) In-scope targets:- Specific systems allowed for testing
2) Time window: mention testing dates & times
3) Permitted Techniques: list allowed
4) Prohibited Techniques: Avoid destructive activities
5) Point of contact: Provide responsible Person's details

b) Foot Printing is the Process of collecting information about a target organization.

Types of Foot Printing

1. Passive Foot Printing: collects information from Public sources without directly interaction with the target
2. Active Foot Printing: collects information by directly interaction with target systems.

Tools used:

1) WHOIS: Finds domain registration details
2) nslookup/dig: Finds IP addresses and DNS record.
3. search Engin operators:
   - ⇒ site: retail10.example: - Finds website page.
   - ⇒ Filetypes: Pdf - Find Pdf
   - ⇒ intitle: login - Finds login page